FISH-SIM-V0 · RESEARCH REPORT

Robot Fish Simulator Survey

Six simulation tools applied to one soft, hydraulically actuated robot fish: what each one takes as input, what it produces, how it works inside, and what it can and cannot answer.

Branches: main · stonefish · pyelastica · sofa · openfoam · openmodelica Version: 9 Oct 2026

Language: EN (PL version available)

Read the numbers as mechanisms, not as a ranking. Every physical parameter in the six demos is a placeholder that has not been identified on a real robot. The demos also use different parameter sets (for example, the SOFA chamber holds 91 ml and the MuJoCo chamber 30 ml). A speed of 22 cm/s in one tool and 44 cm/s in another says which physics each tool includes, not which tool is more accurate.

1 · Summary

Each tool answers a different question about the same fish. Three tools simulate the whole fish swimming freely, two look closely at the soft tail, and one models the drive train from the battery to the fin.

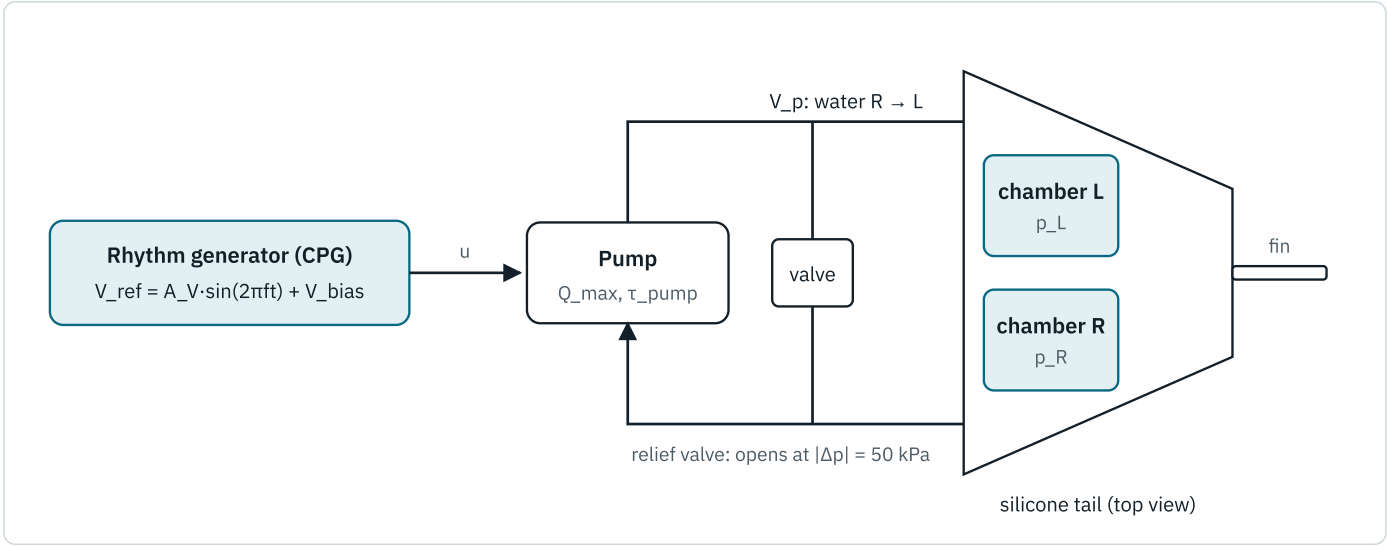
TOOL	BRANCH / FOLDER	LEVEL OF DETAIL	WHAT IT SHOWS IN THIS REPO
MuJoCo	`main` · `MuJoCo/`	Whole fish, rigid segments	Swimming, turning, depth control with a bladder; interactive keyboard control
Stonefish	`stonefish` · `Stonefish/`	Whole fish, rigid segments, marine environment	Hydrodynamics from the mesh, VBS ballast, noisy pressure and IMU sensors, sea current
PyElastica	`pyelastica` · `fish_elastica_demo/`	Whole fish as one soft rod	Chamber-driven bending, free 3D swimming, effect of stiffness on speed, depth control
SOFA + SoftRobots	`sofa` · `SOFA/`	Tail only, 3D finite elements	How the silicone tail and its chambers deform: p–V curve, bend angle, flapping in air and water
OpenFOAM + preCICE + CalculiX	`openfoam` · `OpenFOAM/`	Tail and water flow, 2D	Two-way fluid–structure interaction: vortex wake, added mass, tethered thrust
OpenModelica	`openmodelica` · `FishRobot/`	System model, lumped	Battery → motor → pump → pipes → chambers → tail; energy budget, ballast, FMU export, calibration

Which tool for which question

QUESTION	FIRST CHOICE	ALSO USEFUL
Does the fish swim, turn and hold depth? I want to tune controllers interactively.	MuJoCo	Stonefish
Does the control software work with noisy sensors and a sea current? Path to ROS 2.	Stonefish	MuJoCo
How does body and tail stiffness change speed, cheaply and with a continuous body?	PyElastica	SOFA
What pressure does a chamber need for a given volume and bend? Which wall, fiber or spine design bends best?	SOFA	OpenFOAM (CalculiX part)
What does the water really do: wake, added mass, real thrust or drag?	OpenFOAM FSI	none of the others resolves flow
Are the motor, pump and battery sized right? How long does the battery last? Where is the energy lost?	OpenModelica	none of the others has electrics
How do I turn bench measurements into model parameters?	OpenModelica (`calibrate.py`)	calibration plans in the MuJoCo and SOFA READMEs

2 · The common fish model

All six demos model the same robot concept, a fish with a soft silicone tail bent by water pumped between two chambers. Knowing this loop makes the input tables in section 3 readable.



The shared hydraulic loop. The pump moves a volume V_p from chamber R to chamber L in a closed circuit; the pressure difference Δp = p_L − p_R bends the tail. The rhythm generator commands a volume (not a flow), and an offset V_bias makes the fish turn. Depth is controlled separately: a PID sets the volume of a ballast bladder (MuJoCo, PyElastica), a VBS tank (Stonefish) or a syringe (OpenModelica).

The same quantities, set differently in each demo

QUANTITY	MUJOCO	STONEFISH	PYELASTICA	SOFA	OPENFOAM	OPENMODELICA
Body	ellipsoid hull 25×10×10 cm	same as MuJoCo	whole fish = rod 0.40 m	none (tail clamped)	2D head, half-ellipse 60 mm, fixed	lumped mass 1 kg
Tail	5 rigid segments × 40 mm, 3 actuated	5 rigid segments × 40 mm	continuous, chamber zone 0.20–0.32 m	3D FEM, 0.20 m	2D FEM, 0.15 m	1 rotational DOF
Pump flow Q_max	60 ml/s	60 ml/s	160 ml/s	250 ml/s	none (pressure prescribed)	0.3 ml per revolution
Volume amplitude A_V	8 ml	8 ml	8 ml	17 ml	P0 = 15 kPa	CPG duty A = 0.8
Relief pressure	50 kPa	50 kPa	50 kPa	50 kPa	none	50 kPa
Ballast	bladder 0–40 ml	VBS 0.5–40.5 ml	bladder 0–40 ml	none	none	syringe 12.5 ml

3 · The six tools

Each tool section has the same parts: a fact sheet, the working principle, the inputs (what the demo uses and what the tool could also take), the outputs with example values from the repo, the demo stages, and the limitations. Screenshots and videos come from the programs themselves, recorded for this report on 9 Oct 2026; where a tool has no GUI, the demo's own animation is shown.

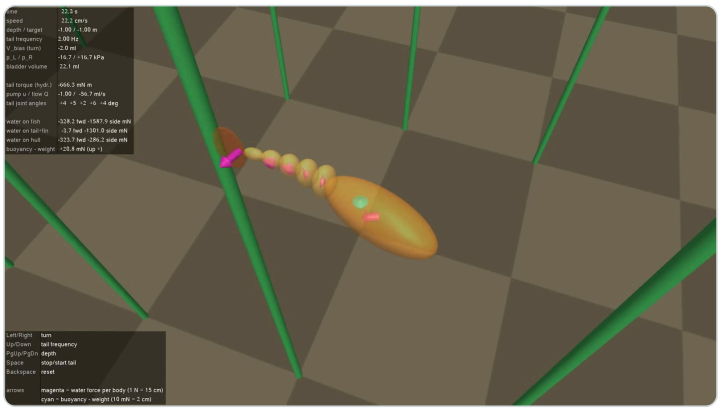
WHOLE FISH · BRANCH MAIN

● MuJoCo

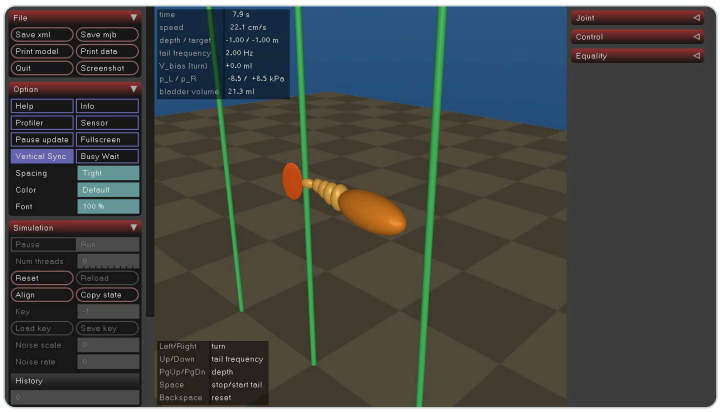
A fast rigid-body engine with a simple built-in water model. The fish is a hull and a chain of five rigid tail segments; buoyancy and hydraulics are added in Python.

METHOD	INTERFACE	TESTED VERSION
Rigid multibody dynamics + quasi-static ellipsoid fluid forces	Python bindings, MJCF model generated in code	MuJoCo 3.15.0, Python ≥ 3.10
INSTALL	**SPEED**	**LICENCE***
`pip install -r requirements.txt`, minutes	Live in the interactive viewer; scenarios of 10–130 s run headless	Apache-2.0

In the program: GUI and demo video



VIDEO MuJoCo viewer recorded live at 2560×1440 and slowed to 0.5× (30 s of simulation in 61 s). The overlay shows the state (time, speed, depth, tail frequency, V_bias, chamber pressures, bladder volume), the drive (hydraulic torque on the tail, pump command and flow, tail joint angles) and the water forces on the whole fish, on the tail with the fin, and on the hull, split into forward and sideways parts in mN. Magenta arrows show the water force on each body and the cyan arrow shows buoyancy − weight; the bodies are drawn transparent so the arrows stay visible. MuJoCo reports the fluid force only as a total, so the split per body is recomputed on a copy of the state (`fishsim/readouts.py`). A script presses the keys at these simulation times: → at 8 s, ← at 17 s, → at 26 s, PgDn at 28 s (dive). Video file: `docs/media/mujoco_demo.mp4` (plays in the HTML version).



SCREEN The full MuJoCo viewer: File, Option and Simulation panels on the left, Joint, Control and Equality on the right, the demo's status overlay in the scene.

How it works

- **Body.** The hull is an ellipsoid on a free joint. The tail is five ellipsoids on hinge joints with springs and dampers, a pseudo-rigid-body approximation of the silicone. The caudal fin is a flat ellipsoid on the last segment.
- **Water.** MuJoCo's built-in ellipsoid model gives blunt and slender drag, angular drag, Kutta lift and Magnus force for each part. There is no flow field.
- **Added mass, corrected.** MuJoCo applies only the gyroscopic part of the added-mass force: the source (`engine_passive.c`) omits the acceleration term that the docs show. On its own that part pushed the fish backwards (4.5 cm/s instead of 22 cm/s). The demo switches it off and adds the added mass as joint `armature` instead.
- **Buoyancy.** MuJoCo does not compute buoyancy. The demo applies it to each body, with the centre of buoyancy fixed in the hull frame so that tilting produces a righting moment. The hull mass is solved for neutral buoyancy at a half-full bladder.
- **Hydraulics.** A Python model runs before every physics step: a first-order pump, $\Delta p = (V_p - A \cdot r \cdot L)/C_h$ clipped at ±50 kPa, and a tendon force $F = A \cdot r \cdot \Delta p$ spread over the three actuated joints with weights from 1 to 0.4.
- **Control.** A volume rhythm generator for the tail and a cascade depth PID that sets the bladder volume, followed by a rate-limited ballast pump.

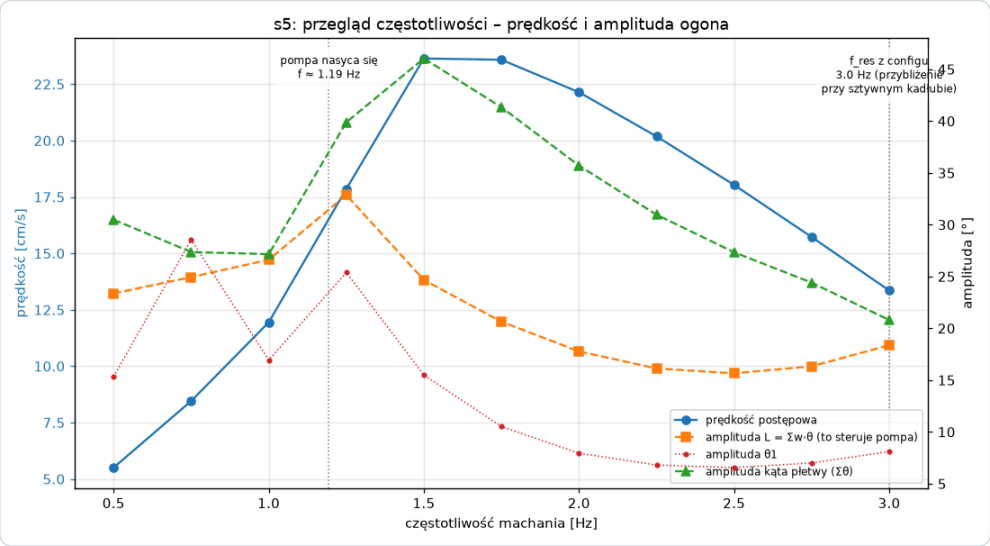
Inputs

CATEGORY	USED IN THE DEMO (`FISHSIM/CONFIG.PY`)	ALSO POSSIBLE WITH MUJOCO
Environment	water ρ 1000 kg/m³, μ 0.0009 Pa·s, step 2 ms (`implicitfast`), floor at −3 m	height fields, terrain meshes
Geometry	hull ellipsoid 0.25 × 0.10 × 0.10 m; 5 tail segments × 40 mm tapering to 0.4; fin 70 × 6 × 120 mm	MJCF or URDF files, STL/OBJ meshes

CATEGORY	USED IN THE DEMO (`FISHSIM/CONFIG.PY`)	ALSO POSSIBLE WITH MUJOCO
Joints	±35°, actuated stiffness 0.3 N·m/rad, passive stiffness from a 3 Hz resonance, damping 0.005 N·m·s/rad	any joint type, tendons, equality constraints
Water model	`fluidcoef` = blunt 0.5, slender 0.25, angular 1.5, Kutta 1.0, Magnus 1.0; added-mass mode	deformable `flex` bodies (observed: the fluid model ignored them; the docs are silent)
Hydraulics	A_eff 4·10⁻⁴ m², r_eff 0.05 m, C_h 8·10⁻¹¹ m³/Pa, p_max 50 kPa, chamber 30 ml, Q_max 60 ml/s, τ_pump 30 ms	any Python model coupled through `ctrl`
Ballast and control	bladder 0–40 ml at 10 ml/s; depth PID gains; tail 2 Hz, A_V 8 ml, V_bias, K_v 10 s⁻¹	sensors (IMU, force, touch), cameras, GPU batches (MJX) for reinforcement learning
Live input	viewer keys: ←/→ V_bias ±0.5 ml, ↑/↓ frequency ±0.25 Hz, PgUp/PgDn depth ±0.25 m, Space start/stop	any Python callback

Outputs

OUTPUT	CONTENT	EXAMPLE VALUE
s1 straight swim	CSV + PNG: speed, five joint angles, p_L, p_R, Δp, tendon force, pump command	≈22 cm/s at 2 Hz after ≈2 s
s2 water on/off	speed with water, without water, with MuJoCo's default added mass	22 / 0 / 4.5 cm/s
s3 turning	XY trajectories for V_bias from −3 to +3 ml	±3 ml: 6.3°/s, radius ≈1.6 m
s4 depth steps	depth, setpoint, bladder volume, pitch	1 m in ≈35 s swimming, ≈6 s hovering; overshoot ≈3 cm
s5 frequency sweep	speed, amplitudes, pump saturation for 0.5–3 Hz	peak 23.6 cm/s at 1.5 Hz
s6 righting	roll, pitch and depth after a 30° roll	upright in ≈0.3 s
Model report	text: masses, volumes, net buoyancy, CG/CB, stability margins of the time step	hover drift 0.000 cm in 5 s



Stage 6 frequency sweep (plot labels are in Polish on the main branch). Speed peaks at 1.5 Hz, just above 1.19 Hz, where the pump starts to saturate.

Demo stages

`0` `check_api.py`: checks the API; a ball with the density of water sinks, which shows that MuJoCo has no built-in buoyancy

`1` `show_model.py`: static model and mass/buoyancy report

`2–5` `run_scenarios.py`: hover and righting, offline hydraulics, swimming with the added-mass diagnosis, turning, depth steps

`6` `sweep_frequency.py`: frequency sweep

`7` `run_viewer.py`: interactive control from the keyboard

Limitations

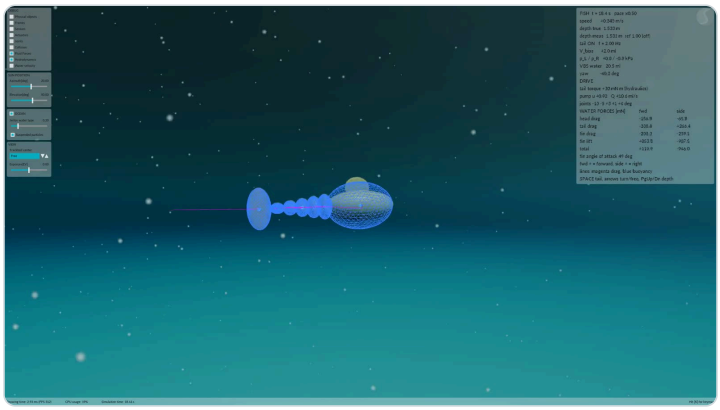
- No flow simulation: no wake, no vortices, no interaction between body parts through the water.
- Diagonal added mass, computed once for a straight tail, so the fish accelerates too quickly.
- The tail is rigid segments, not a continuum. The hydraulics are linear (constant C_h, ideal valve, no losses).
- No waves or free surface; buoyancy is faded out over ±5 cm near the surface as a patch.

Stonefish

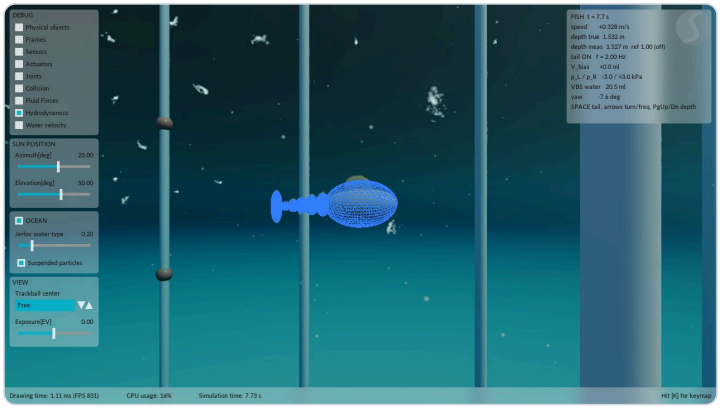
A marine-robotics simulator that computes water forces from the actual surface mesh and ships sensors, ballast tanks and ocean currents. The fish has the same layout as in MuJoCo; the extra physics is written in C++.

METHOD	INTERFACE	TESTED VERSION
Bullet multibody (Featherstone) + per-triangle hydrodynamics	C++20 application + XML scenarios + JSON config	Stonefish 1.5.0, built from source
INSTALL	**SPEED**	**LICENCE***
CMake build; OpenGL ≥ 4.3, SDL2, Freetype, GLM, OpenMP	5–8× faster than real time in console mode at 2000 Hz; GUI ≈800 FPS	GPL-3.0

In the program: GUI and demo video



VIDEO Stonefish GUI recorded live at 2560×1440: scenario s3 (turn with V_bias = +2 ml) slowed to 0.5×, so 25 s of simulation fill 50 s. The panel on the right adds the drive (hydraulic torque, pump, joint angles) and the water forces in the head frame: drag on the head, the tail and the fin, the fin lift from the demo's FinLift model, the total, and the fin angle of attack. The library's "Fluid Forces" lines are on (magenta = drag, blue = buoyancy). From 30 s to 40 s of the video the "Hydrodynamics" view is on as well. Video file: `docs/media/stonefish_demo.mp4` (plays in the HTML version).



SCREEN Debug view "Hydrodynamics": the shapes Stonefish uses for buoyancy, drag and added mass, drawn in blue. The status panel (top right) is part of the demo.

How it works

- **Body.** Head plus five tail links on revolute joints and a fin. The meshes are generated by `make_meshes.py` (320 triangles for physics, 5120 for display), which also balances the mass and sizes the steel ballast.
- **Water from geometry.** Buoyancy is computed from hydrostatic pressure on every triangle. Pressure drag acts per face, skin friction acts along the surface, and added mass comes from an automatically fitted sphere, cylinder or ellipsoid.
- **Fixes in C++.** Face drag cannot produce lift, so `FinLift.cpp` adds a flat-plate lift on the fin. Without it the fish reaches only 2.7 cm/s instead of 34 cm/s. The library's skin friction was about 250× too high and is replaced by a Blasius law.

- **Drive.** `Hydraulics.cpp` holds the same lumped pump and chamber model as MuJoCo. `TailDriver.cpp` applies joint torques τ_i = w_i·F − k_i·θ_i as an internal drive (action equals reaction).
- **Sensors and control.** The pressure sensor (50 Hz, σ 50 Pa) and the IMU (100 Hz) are noisy. The depth PID reads the filtered pressure and sets the VBS water volume; a heading PI on the IMU yaw sets V_bias.
- **Ocean.** Seabed, posts for visual reference, water type and a uniform current.

Inputs

CATEGORY	USED IN THE DEMO	ALSO POSSIBLE WITH STONEFISH
Config file	`config/default.json`, per-scenario patches `sN_*.json`, single overrides `--set key=value` (no recompilation)	any parameter exposed by the application
Scenario XML	`fish_base.scn`: links, joints, motors, VBS, pressure sensor, IMU, encoders; `world_base.scn`: seabed, ocean, current	geometric waves (GPU-computed; body interaction still experimental), other current types, custom hydrodynamic coefficients
Geometry and materials	OBJ meshes; hull density 600, silicone 1100, steel ballast 7850 kg/m³ (sized automatically)	any OBJ geometry
Hydraulics and rhythm	same values as MuJoCo: Q_max 60 ml/s, A_V 8 ml, f 2 Hz, p_max 50 kPa; fin lift slope 2.8/rad	thrusters, rudders, servos
Sensors	pressure 50 Hz with σ 50 Pa; IMU 100 Hz with angle, gyro and accelerometer noise	cameras, sonar, DVL, GPS, thermal camera
Control and live input	depth setpoint schedule, heading PI; GUI keys for the CPG, V_bias, frequency and depth	ROS 2 interface (`stonefish_ros2`), not used here
Simulation	2000 steps/s (2 ms was unstable), hydrodynamics every step, log every 20 steps	real-time or batch mode

Outputs

OUTPUT	CONTENT	EXAMPLE VALUE
CSV log, 100 Hz	≈70 columns: pose, velocities, sensor readings, joint θ/ω/τ, pump, chambers, VBS, forces per body part	full logs in `results/logs/`
s1 hover / righting	depth drift; roll after a 30° tilt	0.21 mm in 10 s; 30° → 10° in 5 s
s2 swim	speed with and without fin lift, locked tail; force balance	34 cm/s; 2.7 cm/s without lift; fin lift 0.51 N
s3 turning	turning radius for V_bias ±1/±2/±3 ml	20.4 / 10.1 / 6.6 m
s4 depth	true vs measured depth with the noisy sensor	1 → 3 m: overshoot 3.3 cm, within ±5 cm after 21 s
s5 current	drift in a 5 cm/s side current, with and without heading control	heading held at +2.9°, yet 1 m sideways drift in 30 s
s6 sweep, GUI	speed vs frequency; `summary.json`; 3D window	peak 33.9 cm/s at 2 Hz



The Stonefish GUI during scenario s2: the panel on the right shows live speed, true and measured depth, chamber pressures and VBS volume.

Demo stages

1–2 Build Stonefish; generate meshes and the mass balance (`tools/make_meshes.py`)

3 s1: hovering and righting

4 s2: joints and hydraulics, diagnosing where thrust comes from

5 s3–s5: turning, depth on the noisy sensor, side current

6 s6: frequency sweep; `t_internal` checks that the drive is internal

7 ROS 2: skipped

Limitations

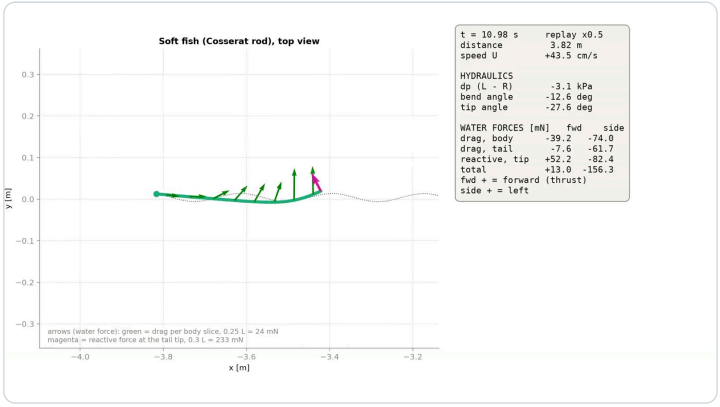
- Rigid-segment tail; quasi-static hydrodynamics with no wake, no delayed stall and no interaction between bodies.
- Added mass averaged over the axes, with no added inertia about the roll axis; friction linearised at 0.2 m/s.
- The GUI needs a GPU. Eight library pitfalls of version 1.5 are documented together with their workarounds.

PyElastica

The whole fish is one soft Cosserat rod: a centreline with a rotating frame that can bend, twist, shear and stretch. It is the cheapest way here to simulate a continuously deforming body that swims freely.

METHOD	INTERFACE	TESTED VERSION
Cosserat rod theory, explicit symplectic Position Verlet	Pure Python, `import elastica as ea`, numba kernels	PyElastica 1.0.0, numba 0.68, Python 3.14 (works; PyElastica officially lists 3.10–3.13)
INSTALL	**SPEED**	**LICENCE***
`pip install -r requirements.txt`, minutes	≈27 µs per step of a 100-element rod; 9–70 s per stage	MIT

In the program: GUI and demo video



VIDEO Free swimming at 2 Hz, top view: 15 s of simulation at 0.5× (30 s), rendered at 2560×1440 by `scripts/animate.py` on the pyelastica branch. PyElastica has no GUI, so matplotlib draws the saved rod positions and the readouts saved in stage 4. These are the speed, Δp, the bend and tip angles, and the water forces in the fish frame: drag on the whole body and on the tail half, the Lighthill reactive force at the tip, and the total. Green arrows show the drag on each body slice; the magenta arrow shows the reactive force at the tip, on its own scale. Video file: `docs/media/pyelastica_free_swim.mp4` (plays in the HTML version).

How it works

- **Body.** A 0.40 m rod with 50 elements. Each element has an elliptical section (from a table of 7 stations), so the tail is soft sideways and stiff vertically. The head is 20× stiffer.
- **Actuation from inside.** The chamber pressure changes the rest curvature of the chamber zone (0.20–0.32 m): $\kappa_{rest} = -A_r \cdot \Delta p / B$. The drive produces no net force; the centre of mass drifts only 10^{-15} m in vacuum.
- **Hydraulics.** The same lumped pump, chamber and valve model as in MuJoCo and SOFA, stepped every 0.2 ms and coupled through the bend angle of the chamber zone.
- **Water.** A custom force: quadratic Morison drag per unit length, plus the Lighthill large-amplitude reactive force at the trailing edge. The built-in Stokes slender-body model is kept only to show it is wrong at this Reynolds number (≈$1.8 \cdot 10^5$): its force is ≈400× too small.
- **Buoyancy and depth.** Per-element weight and buoyancy, a righting moment from the centre of gravity sitting below the axis, a bladder, and a cascade depth PID ported from MuJoCo.
- **Time step.** ≈19 µs, from the explicit stability limit of the rod.

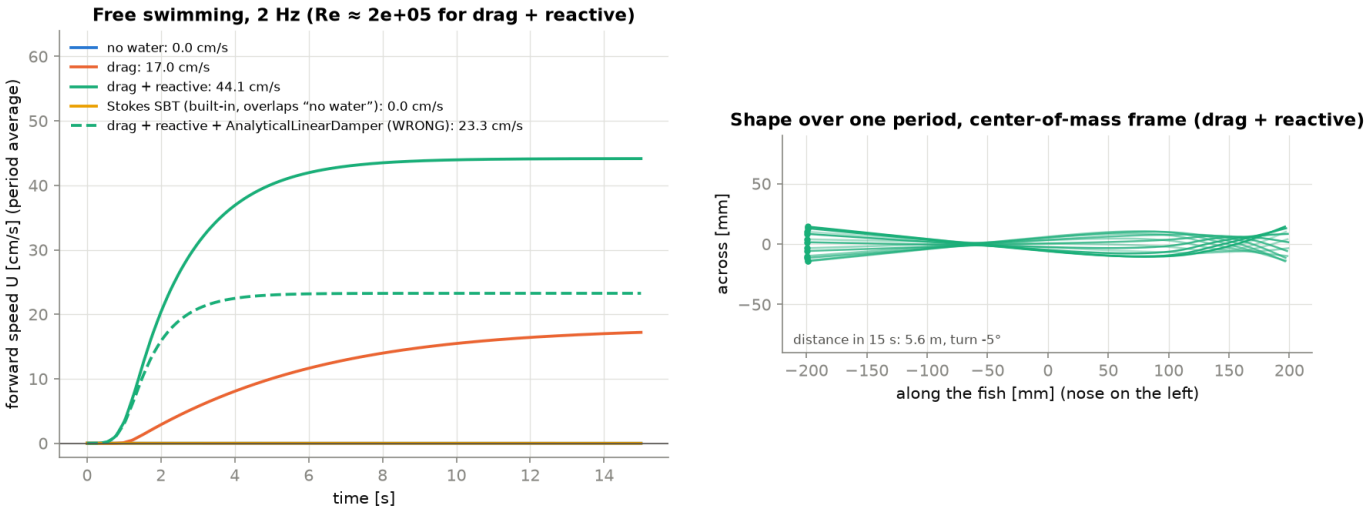
Inputs

CATEGORY	USED IN THE DEMO (`FISHROD/CONFIG.PY`)	ALSO POSSIBLE WITH PYELASTICA
Rod	L 0.40 m, ρ 1050 kg/m³, E 0.5 MPa, ν 0.5, 50 elements	several rods joined by joints, rigid bodies
Shape	half-width and half-height at 7 stations (24 × 34 mm at mid-body), ellipse or circle, stiff head	any per-element radius and stiffness
Hydraulics	A_{eff} $4.3 \cdot 10^{-4}$ m², r_{eff} 5.5 mm, C_h $4.5 \cdot 10^{-10}$ m³/Pa, p_{max} 50 kPa, Q_{max} 160 ml/s, f 2 Hz, A_V 8 ml	built-in muscle torques, uniform torques
Water	model: none, drag, drag+reactive, stokes_sbt; C_n 1.2, C_t 0.01	coupling to an immersed-boundary flow solver (SophT)
Buoyancy and control	CG 5 mm below the axis, bladder 0–40 ml at 10 ml/s, PID at 100 Hz, depth schedule	contact and friction with a floor

CATEGORY	USED IN THE DEMO (`FISHROD/CONFIG.PY`)	ALSO POSSIBLE WITH PYELASTICA
Stage settings	sweep grids (f 0.5–3 Hz × E ×0.5/1/2), durations, damping type	other boundary conditions and dampers

Outputs

OUTPUT	CONTENT	EXAMPLE VALUE
e1 validation	cantilever vs theory, natural frequency, energy drift for n = 25/50/100	n = 50: tip error 1.05 %, ω_1 error 1.8 %
e2 actuation	bend vs pressure and vs volume; internal-drive check	8 ml → 15.8 kPa, 0.377 rad
e3 tethered	tip amplitude, phase, thrust and lateral force for each water model	thrust 149 mN with the reactive force, 5.9 mN with drag only
e4 free swim	speed vs time; body shape over one period; `free_swim.gif`	44.1 cm/s (1.1 body lengths/s); 17 cm/s with drag only
e5 sweep f × E	speed, Strouhal number, tail natural frequency	softest tail at 3 Hz: 55 cm/s; f_n 1.89/2.60/3.64 Hz
e6 depth	depth, bladder volume, pitch, roll, speed	1.0 → 1.5 m settles in 9.9 s, overshoot 23 cm



Stage 4: the choice of water model decides the result. Without the reactive (Lighthill) force the fish reaches 17 cm/s; the Stokes model gives zero.

Demo stages

1 Rod validation against beam theory, convergence in 1/n

2 Rest-curvature actuation and hydraulic coupling; internal-drive check in vacuum

3 Tethered tail with three water models; thrust from momentum balance

4 Free planar swimming, comparison of water models and damping

5 Frequency × stiffness sweep, resonance, Strouhal number

6 3D with gravity, buoyancy, bladder and depth PID while swimming

7 Stiffness export to MuJoCo: not done

Limitations

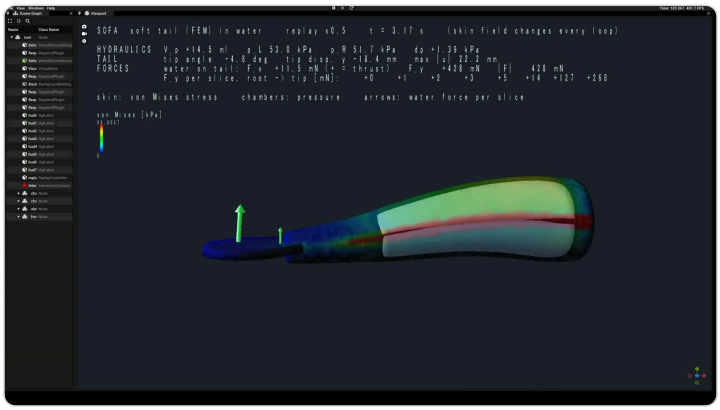
- Local water model with no wake: the Strouhal number is too low (0.12–0.2) and speed is overestimated.
- No distributed added mass, so the natural frequency in water is too high. The section is an ellipse, not a thin fin plate.
- The chamber is reduced to A_r·Δp and C_h; elasticity is linear. A very stiff hydraulic spring breaks the explicit coupling.

SOFA + SoftRobots

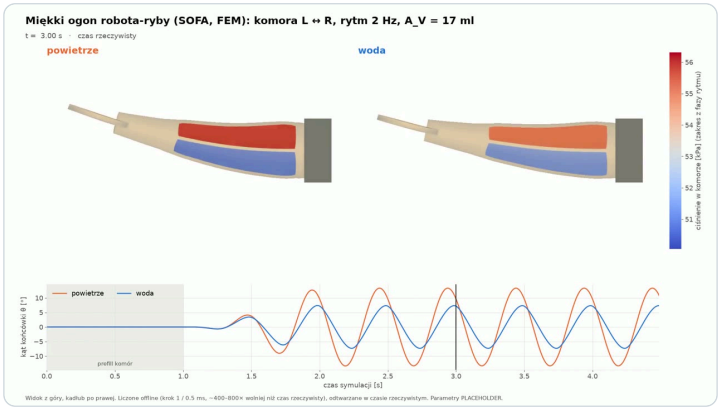
A 3D finite-element model of the silicone tail with real chamber cavities. The pump prescribes the chamber volume and SOFA computes the pressure needed. The tail is clamped at the root, as on a thrust test rig.

METHOD	INTERFACE	TESTED VERSION
Corotational FEM on tetrahedra + Lagrange constraints for chambers	SofaPython3 scenes (`createScene`), C++ engine	SOFA v26.06.00 binary (SoftRobots included)
INSTALL	**SPEED**	**LICENCE***
`scripts/setup.sh`: ≈2 min + 230 MB, builds the CHOLMOD plugin	100–800× slower than real time (coarse mesh, one core)	LGPL-2.1+ (SOFA), LGPL-3.0 (SoftRobots)

In the program: GUI and demo video



VIDEO SOFA GUI (ImGui) at 2560×1440 replaying the water run at 0.5×; the run itself was computed offline (`scripts/record.py`, ≈30 min). The text shows the hydraulics (V_p, p_L, p_R, Δp), the tail (tip angle, tip displacement, max |u|) and the water force on the tail, in total and per slice. Green arrows draw the force on each slice, and the chambers are coloured by pressure. The skin colour changes every loop, each with its own legend: displacement [mm], water load [Pa], then von Mises stress [kPa]. The stress is computed from the recorded node positions with the same corotational model as the FEM. Video file: `docs/media/sofa_gui_replay.mp4` (plays in the HTML version).



VIDEO Rendered video from the sofa branch (PyVista, 6.5 s): air and water side by side with the tip angle below, first in real time, then the last cycle 4× slower. Labels are in Polish. Video file: `docs/media/sofa_flapping.mp4` (plays in the HTML version).



SCREEN SOFA's ImGui interface: scene graph (components of the tail scene) on the left, viewport on the right.

How it works

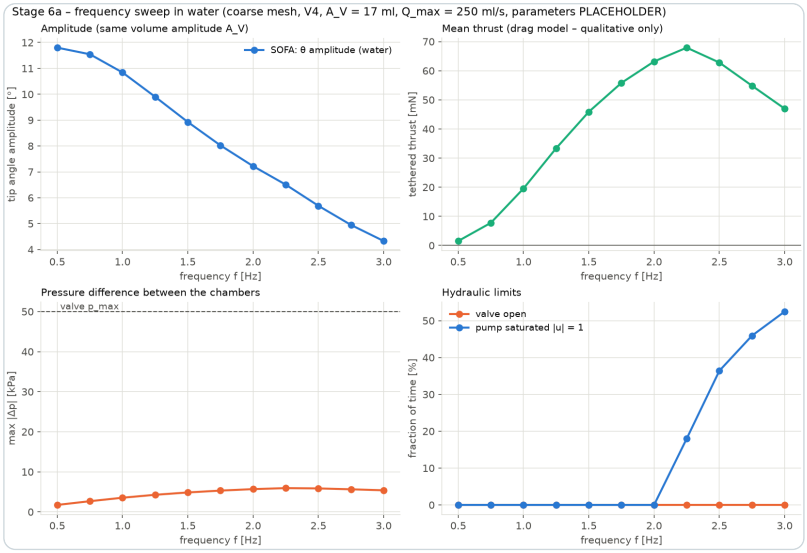
- **Mesh.** gmsh builds the tail with two cavities. The coarse level has 28 112 tetrahedra and 7 744 nodes; the fine level has 142 096 tetrahedra.
- **Material.** Corotational linear elasticity (large rotations, small strains), E 0.3 MPa, v 0.45. The default design V4 adds a spine 20× stiffer in the septum and tension-only hoop fibers every 4 mm.
- **Chambers.** `SurfacePressureConstraint` in volume-growth mode: the pump sets the volume, and the constraint solver returns the pressure.
- **Solver.** Implicit Euler, a Gauss–Seidel constraint solver and the CHOLMOD sparse Cholesky factorisation (4–15× faster than LDL). Static curves are found pseudo-statically, holding each point until the motion dies out.
- **Water.** Custom quadratic normal drag and tangential friction on every skin triangle, applied explicitly; the silicone weight is reduced by buoyancy. There is no added mass, lift or wake.
- **Hydraulics.** A port of the MuJoCo rhythm with pump-lag compensation (without it the volume overshoots by 16 % at 2 Hz), a closed circuit and a 20 ml prefill.

Inputs

CATEGORY	USED IN THE DEMO (`FISHSOFA/CONFIG.PY`)	ALSO POSSIBLE WITH SOFA
Geometry	tail 0.20 m, section 60 × 80 mm tapering to 0.4, fin 70 × 120 × 6 mm; two chambers 0.12 m long, 4 mm wall, 4 mm septum (91 ml each)	any Gmsh, STL, OBJ or VTK mesh; XML `.scn` scenes
Material	E $3{\cdot}10^5$ Pa, v 0.45, ρ 1100; spine E × 20; fibers $2{\cdot}10^5$ N/m every 4 mm	hyperelastic materials, beams (BeamAdapter)
Actuation	f 2 Hz, A_V 17 ml, prefill 20 ml, Q_max 250 ml/s, p_max 50 kPa	pressure control, cable actuators, inverse control (SoftRobots.Inverse)
Environment	air or water; drag coefficients C_n 1.0, C_t 0.01	contact and collision
Numerics	step 2 ms (0.5–1 ms in water), Rayleigh damping, solver cholmod/ldl/cg/warp, mesh level test/coarse/medium/fine	model order reduction (>100× faster per Goury & Duriez 2018; needs offline training)
Run control	environment variables `FISHSOFA_LEVEL`, `_MODE`, `_ENV`, `_WORKERS`	GUI interaction

Outputs

OUTPUT	CONTENT	EXAMPLE VALUE
s1 mesh and sag	mesh quality report; tip sag under own weight on three meshes	−40.5 mm (coarse) to −41.2 mm (fine)
s2a design variants	bend for homogeneous, spine, thick wall, fibers, spine + fibers	1.3° homogeneous vs 31° for V4
s2 p–V curve	pressure and tip angle vs chamber volume, mesh convergence	50 ml: 16.1 kPa and 15.4° (fine mesh)
s3 symmetry	left vs right chamber	error ≤ 0.75 % in angle
s4 flapping in air	angle, pressures, volume conservation, solver iterations	±12.7° at 2 Hz, $\lvert\Delta p\rvert \le 5.7$ kPa
s5 water vs air	amplitude, phase, tethered thrust, lateral force, drag power	±7.3° in water (0.55× air); thrust 66.7 mN
s6 sweeps, video	frequency and Young's modulus sweeps; `flapping.mp4` 1080p60	max thrust 67.9 mN at 2.25 Hz



Stage 6 in water. Amplitude falls steadily with frequency (no resonance in water), thrust peaks at 2.25 Hz, and above 2 Hz the pump saturates. The thrust comes from a drag model and is qualitative.

Demo stages

0	Component check; meaning, sign and units of the chamber volume constraint (≈2.5 min)
1	Meshes and sag under own weight (≈10 min)
2	Design variants V0–V4; p–V and angle curves, mesh convergence (≈35 min)
3	Left/right symmetry (≈3 min)
4	Antagonistic flapping in air (≈25 min)
5	Water vs air, tethered thrust (30–60 min)
6	Frequency and stiffness sweeps, 19 runs (≈1 h on 12 threads)

Limitations

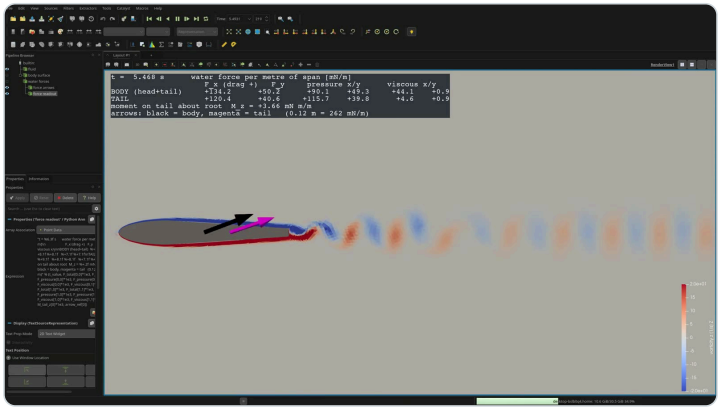
- No flow and no reactive force, so the thrust is qualitative. The tail is clamped; the fish does not swim.
- Linear tetrahedra lock in thin walls, which overestimates pressure by 15–25 %. The material is corotational linear, not hyperelastic.
- Slow: explicit drag limits the step to 0.5 ms in water. In our tests SOFA master (v26.12-dev, unreleased) left a fibre-stiffened chamber vibrating or diverging, so the demo stays on v26.06 (details in `SOFA/README.md`; no upstream issue filed).
- Exporting equivalent joint stiffnesses to MuJoCo (stage 7) was not done.

OpenFOAM + preCICE + CalculiX

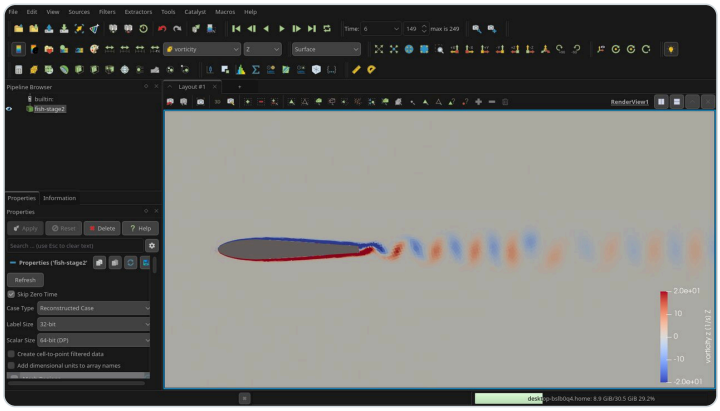
The only demo that solves the actual water flow. A CFD solver and a structural FEM solver exchange forces and displacements every time step until they agree. It is a 2D slice of the tail with the head held fixed.

METHOD	INTERFACE	TESTED VERSIONS
Partitioned FSI: finite volumes + nonlinear FEM, implicit coupling	OpenFOAM dictionaries, CalculiX `.inp`, preCICE XML; Bash and Python tools	OpenFOAM v2606, CalculiX 2.20 (2.23 exists; the preCICE adapter v2.20.2 targets 2.20), preCICE 3.4.1
INSTALL	**SPEED**	**LICENCE***
`setup.sh`, ≈1.5 h, mostly built from source	0.6–1 s per 2.5 ms window; stages 4–6 take 3 h 47 min	GPL-3.0 (OpenFOAM), GPL-2.0+ (CalculiX), LGPL-3.0 (preCICE)

In the program: GUI and demo video



VIDEO ParaView 6.1 at 2560×1440 playing stage 2 (rigid body, U = 0.2 m/s, 10 s of flow) at about 0.5×. `tools/pv_gui.py` adds a text box that follows the animation time, read from the `forces` function objects. It shows the water force on the body and on the tail (total, pressure and viscous parts, in mN per metre of span) and the moment on the tail about its root. Arrows show the total force at the centroids of the body (black) and the tail (magenta). The case was re-run for this report with OpenFOAM v2512 in the official container: 20 665 cells, 90 s on 4 cores. Video file: `docs/media/openfoam_paraview_demo.mp4` (plays in the HTML version).



SCREEN ParaView with the OpenFOAM case loaded: pipeline browser and reader properties on the left, the vorticity field at t = 6 s on the right.

How it works

- **Fluid.** `pimpleFoam` solves incompressible laminar Navier–Stokes in 2D (one cell deep, so forces are per metre of span) on a mesh that deforms with the tail.
- **Solid.** CalculiX runs dynamic nonlinear FEM (large deformations, linear elastic silicone, E 0.3 MPa) of a tail with PneuNet-style chambers.
- **Coupling.** preCICE maps fluid forces onto the solid and solid displacements onto the fluid mesh, and repeats each 2.5 ms window with the IQN-ILS quasi-Newton method until both agree (4.7 iterations on average).

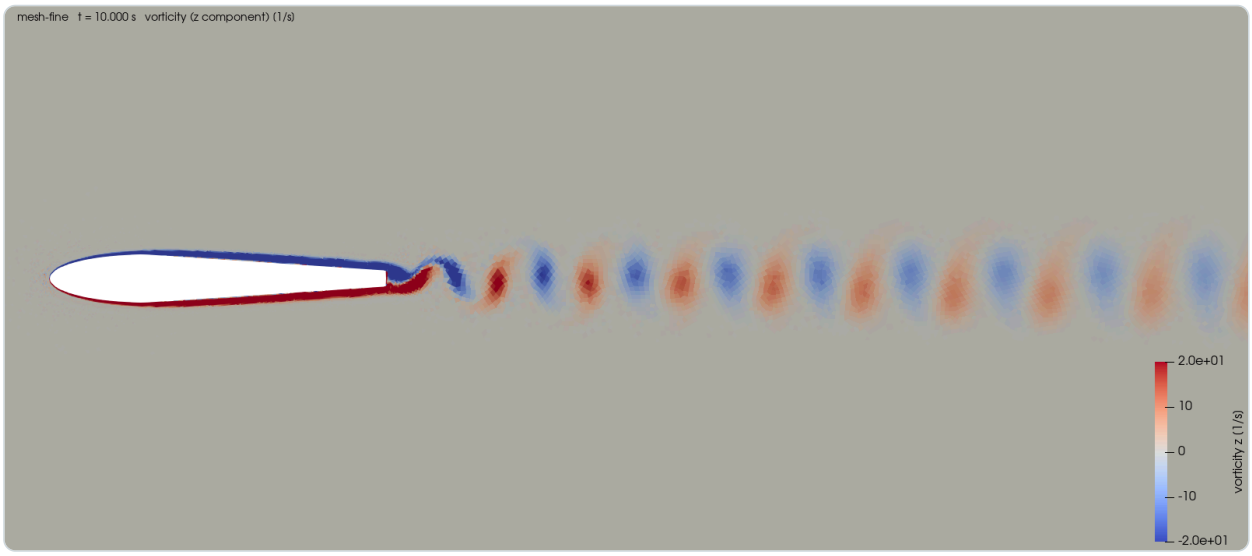
- **Why implicit.** Silicone is almost as dense as water, so explicit coupling is unstable for any time step (the added-mass effect). Stage 3 shows the force growing 13× per window.
- **Actuation.** Prescribed chamber pressure p_L = P0·max(0, sin 2πft), with the opposite half-wave in p_R. Pressure is the input, not volume.
- **Thrust.** The cycle-averaged axial force on the fixed body (head + tail).

Inputs

CATEGORY	USED IN THE DEMO	ALSO POSSIBLE WITH THIS TOOLCHAIN
Geometry (`params.py`)	tail 0.15 m, thickness 30 → 10 mm, chambers at 10–110 mm, 3 mm walls, 10 cells per side; head half-ellipse 60 mm; domain 3L upstream, 8L downstream	full 3D geometry
Materials and water	E $3{\cdot}10^5$ Pa, ν 0.45, ρ 1070 kg/m³; water ρ 1000, ν 10^{-6} m²/s	hyperelastic material, material damping (CalculiX)
Case (`new-fsi-case.sh`)	inflow speed U, time step, end time, P0, frequency, implicit/explicit coupling	free body with a 6-DOF solver
OpenFOAM setup	PIMPLE + GAMG, Laplacian mesh motion, force and vorticity function objects, inlet ramp, slip sides	RANS/LES turbulence, overset or remeshing for large amplitude
CalculiX setup	`fsi.inp`: nonlinear dynamics, HHT α −0.2, PARDISO solver, pressure amplitude tables	contact between chamber walls
preCICE setup	RBF mapping (5 mm radius), IQN-ILS, convergence 10^{-3}, tip watch point	more participants, e.g. a volume-driven hydraulics model
Mesh	gmsh, refinement via `MESH_SCALE` (0.7 / 1.0 / 1.4)	any OpenFOAM mesh generator

Outputs

OUTPUT	CONTENT	EXAMPLE VALUE
Flow fields	velocity, pressure, vorticity, time averages for ParaView	Kármán vortex street behind the body
Forces	pressure and viscous force and moment per patch (head, tail)	rigid body at 0.2 m/s: drag 0.13 N/m, ⅔ pressure
E1 dry tail	tip deflection vs pressure, L/R symmetry	1.2 mm/kPa; 24 mm (16 % of L) at 20 kPa
E3 coupling	explicit vs implicit stability	explicit diverges; implicit stable
E4 actuated	tip amplitude in water vs dry, thrust per cycle	28.3 mm in water vs 20 mm dry at 1 Hz
E5 inflow	mean axial force and Strouhal number vs speed	net drag at 0.05–0.2 m/s; zero only near 0.02 m/s
E6 frequency	amplitude and thrust vs frequency	resonance in water ≈1 Hz (dry 3.39 Hz)



Stage 2, fine mesh, U = 0.2 m/s: the vortex wake that none of the other five tools can show.

Demo stages

0 Official perpendicular-flap tutorial, to validate the install (34 s)

1 Solid only: deflection vs pressure (≈10 s)

2 Fluid only, rigid body: drag and mesh convergence (7 min)

3 Passive tail: explicit vs implicit coupling (10 min)

4 Actuated tail in still water vs dry (16–80 min)

5 Inflow at 0.05/0.1/0.2 m/s (≈30 min each)

6 Frequency sweep 0.5–2 Hz (16–102 min each)

Limitations

- 2D: no tip vortices, and forces are per metre of span. Laminar flow at Re ≈ $4{\cdot}10^4$.

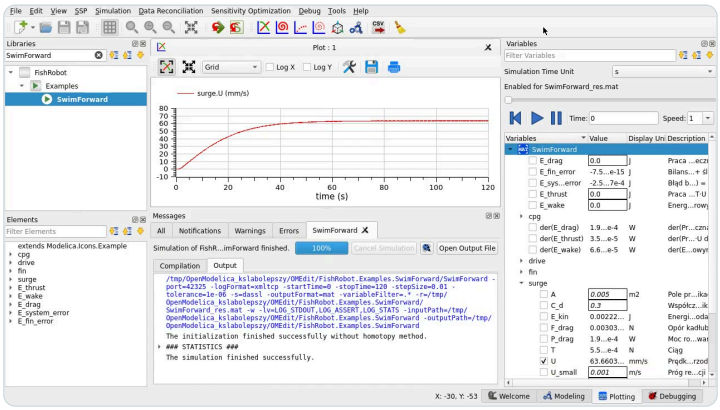
- Fixed head, so no free swimming or recoil. Actuation is by pressure, not volume. No material damping.
- Mesh deformation limits the tip amplitude to ≈15 % of L. Two uniform chambers make a standing wave and a blunt tip, so the tail produces net drag.
- Slow and the most complex of the six to install.

OpenModelica

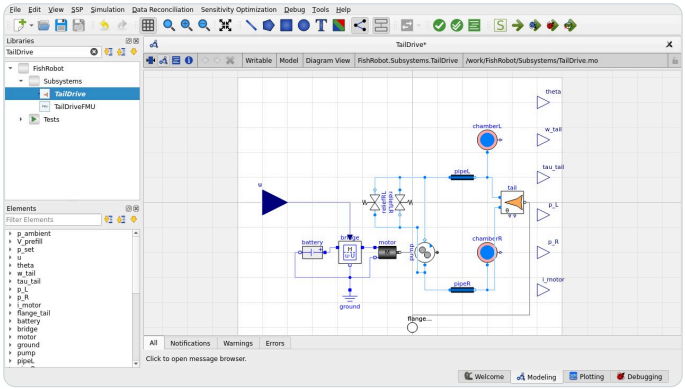
An equation-based model of the whole drive train, from the battery through the motor, pump, pipes and chambers to a one-degree-of-freedom tail. It is the only demo with electrics, an energy budget, FMU export and a full calibration workflow.

METHOD	INTERFACE	TESTED VERSION
Acausal Modelica equations → DAE → C code, variable-step solver	Modelica (OMEdit GUI), Python through OMPython and FMPy	OpenModelica 1.27.1, MSL 4.1.0
INSTALL	**SPEED**	**LICENCE***
apt repository (`install_system.sh`) + venv (`install_user.sh`)	16 simulations ≈0.2 s after ≈1 s of compilation	OSMC-PL 1.8 or AGPL-3.0 (OpenModelica), BSD-3 (MSL)

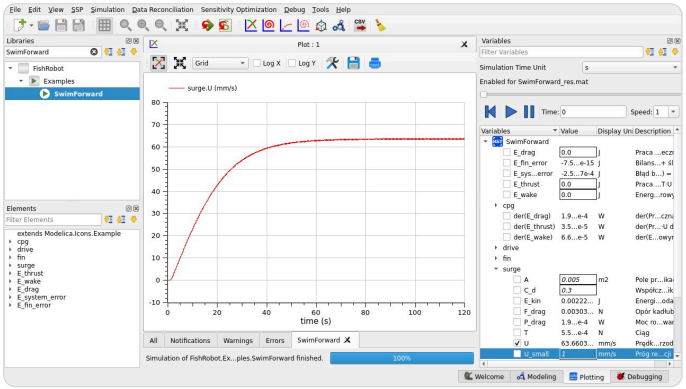
In the program: GUI and demo video



VIDEO OMEdit (OpenModelica 1.27.1, official container): the SwimForward example is simulated (compile and run take ≈5 s; the wait is cut) and the speed U is plotted, then the thrust T is added. Video file: `docs/media/openmodelica_demo.mp4` (plays in the HTML version).



SCREEN OMEdit diagram view of `FishRobot.Subsystems.TailDrive`: battery, H-bridge, motor, pump, pipes, two chambers, relief valves and the tail.



SCREEN OMEdit plotting view after simulating SwimForward: the speed settles at 63.7 mm/s, the same 6.4 cm/s as in the README.

How it works

- **Acausal components.** Parts are joined by physical connectors. The custom hydraulic port carries pressure p and flow V_flow, whose product is power. The compiler sorts the equations itself, so there is no input/output wiring.
- **Drive chain.** 2S LiPo battery 7.4 V → H-bridge → DC motor → gear pump ($Q = D\cdot\omega - k_{leak}\cdot\Delta p$) → pipes (laminar plus Haaland turbulent friction, minor losses) → chambers with a nonlinear p–V table → relief valves.
- **Tail and fin.** The tail is a rotary piston with one degree of freedom: $(J + J_{added})\cdot\ddot{\theta} = \tau - k\theta - c\dot{\theta} - c_h|\dot{\theta}|\dot{\theta}$. Thrust comes from Lighthill's rigid-fin formula and drives a surge equation.
- **Ballast.** A motor and lead screw drive a syringe piston; a heave equation and a cascade depth PID close the loop.
- **Energy.** Every loss is integrated, and the balance closes to $\approx 10^{-6}$.

- **Export and calibration.** `TailDrive` exports as an FMU 2.0 for co-simulation. Ten bench models and `calibrate.py` (least squares) identify the parameters step by step, from the motor to the water.

Inputs

CATEGORY	USED IN THE DEMO	ALSO POSSIBLE WITH MODELICA
Electrical	battery 7.4 V, R_int 0.05 Ω, 16.3 Wh; motor R 1 Ω, L 0.5 mH, k 0.01 N·m/A, rotor inertia	SoC-dependent voltage, PWM losses, thermal models
Hydraulic	pump 0.3 ml/rev, leakage, η 0.8; pipe 0.2 m × 4 mm, ζ 1.5, roughness 25 µm; valve 50 kPa; chamber p–V table (11 points or a CSV file)	`Modelica.Fluid` and media, compressible or two-phase (a variant is already in `HydraulicsFluid/`)
Tail and fin	displacement $2 \cdot 10^{-5}$ m³/rad, k 2 N·m/rad, inertia, added inertia, damping; fin span 0.06 m, C_T 1.0	a p–V curve from SOFA or from measurement
Body and ballast	m 1 kg, added mass, C_d 0.3, area 0.005 m²; syringe 20 mm × 40 mm (12.5 ml), lead 1 mm	multibody library for 3D motion
Control	CPG amplitude 0.8, 1.5 Hz, bias; depth PID gains	any block-diagram controller
Measured data	CSV files for `calibrate.py`: motor step, pump, pipe, chamber cycles, valve, tail, thrust, hull tow and coast-down, ballast	tables from `.mat`/`.txt`, FMU import
Run time	`-override` parameters, OMEdit modifiers	solver choice: DASSL, IDA, CVODE, Euler

Outputs

OUTPUT	CONTENT	EXAMPLE VALUE
Time series (.mat)	every variable: p_L, p_R, flows, currents, motor speed, θ, thrust, speed, depth, ballast volume	valve opens at ≈51 kPa (HydraulicsStep)
Energy budget	loss per component, mechanical output, stored energy, balance error	1.8 W at 1 Hz → ≈9.2 h on 16.3 Wh
Swimming	surge speed, thrust, battery power vs frequency	best 8.5 cm/s at 0.5 Hz; 16.6 W at 4 Hz
Depth control	depth, piston position, current	3 cm/s descent, 6 % overshoot, 5 mm error
FMU	`TailDrive.fmu` (linux64, 1.8 MB) + comparison with OpenModelica	difference 0.016 % of amplitude
Calibration reports	fitted parameters with 1σ uncertainty and correlation matrix	e.g. hull C_d 0.402 ± 0.9 % (synthetic data)



motor winding 78.5 % pipes 9.3 % motor bearings 8.4 % pump friction 2.7 % battery 0.8 % tail 0.1 %

Where the energy goes (EnergyBudget scenario: 60 s at 1 Hz, 1.8 W). With placeholder parameters the motor is oversized, so its winding dominates. In SwimForward only ≈0.01 % of battery energy becomes useful propulsion work. The demo commits no plots; they are generated by the scripts.

Demo stages

1–2 Pipe, chamber and relief valve with tests (laminar, turbulent vs MSL, closed loop)

3–4 Battery, H-bridge, motor, pump; tail, CPG and the TailDrive subsystem; frequency sweep

5–7 Energy budget; ballast and depth control; fin and surge, forward swimming

8–9 FMU export and FMPy loop; `Modelica.Fluid` variant (same results within 0.5 %, 6× slower)

10 Calibration benches and `calibrate.py` for 10 subsystems

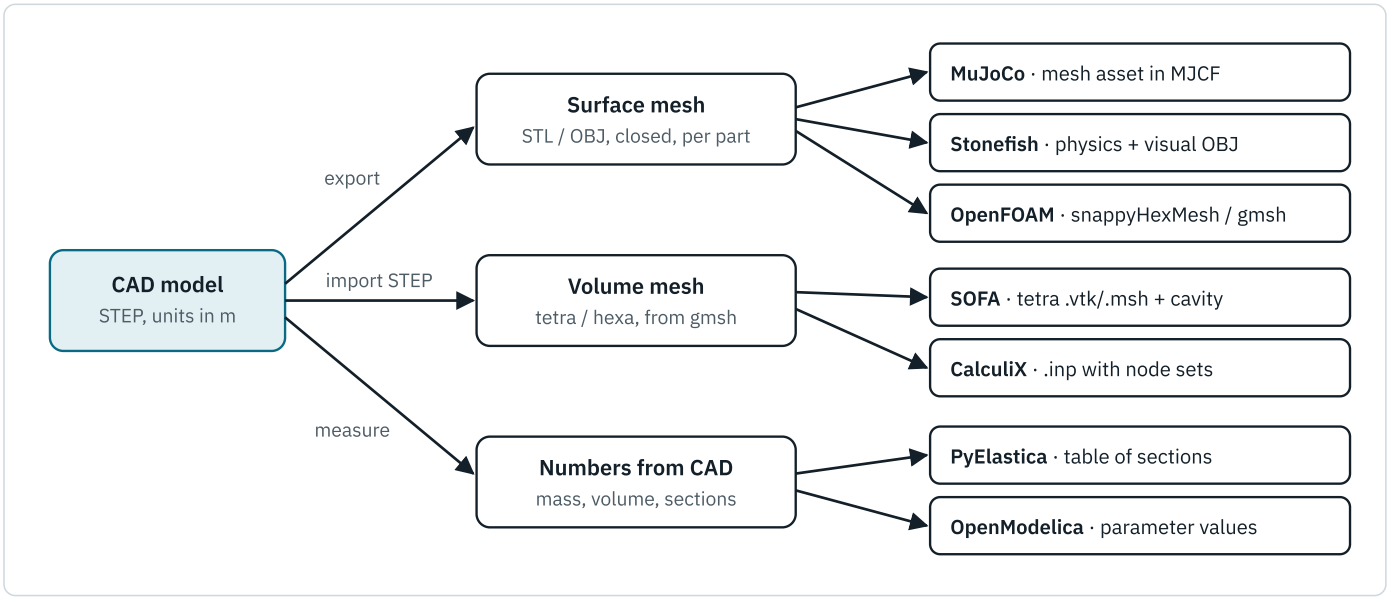
Limitations

- The tail has one degree of freedom (no travelling wave); thrust is a placeholder Lighthill formula.
- Surge, heave and yaw are not coupled; drag and added mass are constants.
- Lumped incompressible hydraulics without hysteresis; ideal electrics. The FMU runs only on the OS it was built on.

4 · From CAD to simulation, and setting parameters

None of the six demos imports a CAD file today. MuJoCo, Stonefish and PyElastica build the fish from simple shapes or a table of sections in code; SOFA and OpenFOAM generate the tail parametrically in gmsh; OpenModelica uses no geometry at all. Once a CAD model of the real robot exists, each tool needs something different from it. This section shows what, how to get it there, and how the physical and numerical parameters are chosen and set.

What each tool needs from the CAD model



One STEP model feeds three kinds of input. Rigid-body tools take surface meshes of each part; the FEM solvers take a volume mesh of the silicone with the chamber cavities; PyElastica and OpenModelica take only numbers measured on the CAD model. The OpenFOAM fluid mesh is built around the body's surface mesh.

CAD import, tool by tool

TOOL	WHAT IT NEEDS	FORMATS	PATH FROM CAD	WATCH OUT FOR
MuJoCo	one rigid body per part (hull, each tail segment, fin), with mass and inertia	STL, OBJ as `<mesh>` assets in MJCF; URDF	Split the tail into segments in CAD, export each part to STL/OBJ in metres, add it as a mesh geom. Mass and inertia come from the mesh and a density, or are typed in from CAD mass properties. A URDF exported from CAD can be loaded and saved as MJCF.	Collisions use the convex hull of each mesh. The fluid model sees only an equivalent ellipsoid per geom, not the mesh shape. Buoyancy still has to be added by the user (as the demo does).
Stonefish	a closed, low-poly physics mesh and a smooth visual mesh per link	OBJ (used here), referenced from the `.scn` XML	Export each link to OBJ, decimate the physics copy to a few hundred triangles (the demo uses 320), keep a fine visual copy. `tools/make_meshes.py` shows the workflow with trimesh + manifold3d: watertight check, volumes, mass balance, ballast sizing, XML filled from templates.	Buoyancy, drag and added mass are computed per triangle, so the mesh must be closed with outward normals. More triangles make every step slower. Materials may be defined only in the main scenario file.
PyElastica	no mesh: a centreline and one cross-section per element	none (numbers in `fishrod/config.py`)	Slice the CAD body at stations along its length (for example `trimesh.section`), read the half-width a(s) and half-height b(s), or compute area and second moments directly, and fill the profile table.	The rod cannot contain cavities. The chambers enter only through A_r (bend per pressure) and C_h (compliance), which are better measured than computed.
SOFA	a tetrahedral volume mesh of the silicone and a closed triangle surface for each chamber cavity	`.msh` (Gmsh), `.vtk` (legacy 4.2), STL/OBJ surfaces	In gmsh replace the parametric geometry of `fishsofa/mesh_gen.py` (already built with the OCC kernel) by `gmsh.model.occ.importShapes("tail.step")`. Define physical groups for the clamped root face and the cavity walls, then mesh.	Use at least 2–3 elements through each thin wall: with 1–2 linear tetrahedra the pressure comes out 15–25 % too high. VTK 5.1 crashes `MeshVTKLoader`. Each cavity surface must be closed.

TOOL	WHAT IT NEEDS	FORMATS	PATH FROM CAD	WATCH OUT FOR
OpenFOAM + CalculiX	fluid: a mesh of the water around the body with named patches; solid: a FEM mesh of the tail with node sets	fluid: STL for snappyHexMesh, or `.msh` via `gmshToFoam` (used here); solid: CalculiX `.inp`	Fluid: export the body surface to STL and mesh with blockMesh + snappyHexMesh (3D), or import STEP into gmsh as in `tools/make_geometry.py`. Solid: mesh in gmsh or FreeCAD FEM and write `.inp` with sets for the clamped root, the wet surface (coupling) and the chamber faces (pressure load).	Patch and set names must match `preciceDict` and `config.yml`. The fluid mesh deforms with the tail, which limits the tip amplitude to ≈15 % of L. A 2D case needs a one-cell-thick extrusion.
OpenModelica	no geometry; lumped values only	none for physics; MultiBody can show STL shapes for visualisation	Read numbers off the CAD model: masses, inertias, centre of buoyancy, chamber volume and its change per radian of bend (D_tail = dV/dθ), pipe lengths and bores, syringe bore and stroke.	The chamber p–V curve depends on the silicone, not only on the shape: take it from SOFA or from a syringe test (the model already reads it from CSV).

How the parameters are chosen

Every value in the demos is a placeholder. Real values come from five sources, used in this order:

1. **Datasheets:** motor R, L, k and rotor inertia, pump displacement, battery voltage and capacity, sensor rates and noise.
2. **CAD mass properties:** masses, volumes, centres of gravity and buoyancy, inertias, chamber volume.
3. **Material tests:** silicone density and Young's modulus (tensile or indentation test of a cast sample).
4. **Bench identification, following the energy flow:** motor → pump → pipe → chamber → valve → tail statics → tail dynamics → thrust → hull → ballast. Each step fixes a few parameters with the previous ones already known. This is the 11-step procedure of the OpenModelica demo (`calibrate.py`), and the same experiments serve the other tools.
5. **Detailed models as references for simple ones:** the SOFA p–V curve fills the OpenModelica chamber table; OpenFOAM tip amplitude, added mass and thrust check the water coefficients of MuJoCo, Stonefish and PyElastica.

Numerical parameters (time step, mesh size, number of elements) are not physics. They are chosen by a stability rule and checked by a convergence test, never tuned to match a result. The Stonefish demo states the rule plainly: when the first version swam at 0.3 cm/s, the cause was diagnosed instead of tuning coefficients until the fish moved.

TOOL	TIME STEP	MESH OR DISCRETISATION
MuJoCo	2 ms, `implicitfast`; stiffest mode $\omega_n \cdot dt \approx 0.05$ (rule: < 0.5)	5 tail segments (from the design, not from convergence)
Stonefish	0.5 ms: explicit joint springs need $\omega \cdot dt < 2$, and 2 ms exploded; 4000 Hz gives the same speed	physics mesh of 320 triangles per body
PyElastica	$dt < \min(\Delta l/c, \Delta l^2/(r \cdot c)) \approx 19$ µs, computed automatically	n = 50 elements: ≈1 % error against beam theory
SOFA	2 ms in air; 0.5 ms in water, from the explicit drag rule $c \cdot dt/m < 0.5$	coarse/medium/fine; coarse overestimates pressure by 20–27 % against fine
OpenFOAM FSI	2.5 ms, identical in OpenFOAM, CalculiX and the preCICE window	three meshes: 1.0 % drag change between the two finest; 1.5× domain changes drag by 3.8 %
OpenModelica	variable step (DASSL), tolerance 10^{-6}	none; `nominal` attributes scale the variables

How the parameters are set, and which experiment identifies them

TOOL	WHERE THEY LIVE	HOW TO CHANGE THEM	COMPUTED AUTOMATICALLY	IDENTIFIED BY (PLAN IN THE README)
MuJoCo	`FishConfig` dataclass, `fishsim/config.py`	edit the file, or `FishConfig(tail_freq=1.5)` in Python; the MJCF model is regenerated	hull mass (neutral at half bladder), fore-aft centre of buoyancy (trim), passive joint stiffness (from a 3 Hz resonance)	tail on a bench (pressure vs angle → A·r, joint k, C_h); tail in water (resonance, damping); tethered thrust; pool video (speed → C_blunt, C_slender; turning → C_Kutta); underwater weighing of the bladder
Stonefish	`config/default.json` + scenario patches `config/sN_*.json`	edit the JSON or pass `--set rhythm.freq=1.5`; no recompilation	steel ballast mass and position, neutral VBS volume, mesh volumes (`make_meshes.py`)	the MuJoCo procedure unchanged, plus roll damping and turning radius
PyElastica	dataclasses in `fishrod/config.py` (Rod, Fish, Hydraulics, Water, Buoyancy, stages)	edit, or `HydraulicsConfig(tail_freq=1.5)` in a script	stable time step, bladder position, section stiffness matrices	bend vs pressure and volume (A_r, C_h, E); natural frequency and damping in air and water; tethered thrust and pool speed (C_n, C_t, reactive coefficient); depth step (bladder pump, PID)
SOFA	`TailConfig` in `fishsofa/config.py`; environment variables `FISHSOFA_*`	edit, e.g. `young_modulus`, `tail_freq`; the mesh is regenerated and cached by a hash of the geometry	meshes, cavity volumes, mass of the water in the chambers	syringe into one chamber in 5 ml steps with a pressure gauge and a top-view photo: fit E to the p–V curve, then the spine factor and hoop stiffness to the angle–volume curve; thrust on a scale in a tub

TOOL	WHERE THEY LIVE	HOW TO CHANGE THEM	COMPUTED AUTOMATICALLY	IDENTIFIED BY (PLAN IN THE README)
OpenFOAM FSI	`tools/params.py` (geometry, materials); `system/caseParams`; dictionaries; `fsi.inp`; `precice-config.xml`	`new-fsi-case.sh --u 0.1 --p0 15000 --freq 1.5 --coupling implicit` writes a new case	meshes (gmsh), actuation pressure tables (`make_actuation.py`)	head on a one-axis load cell in a tank, camera above: mean axial force, tip amplitude dry and in water, inflow speed of zero mean force → E and damping
OpenModelica	`parameter` declarations in the `.mo` files	OMEdit parameter dialog, modifiers such as `cpg(f=1.5)`, or `-override cpg.f=1.5` at run time (OMPython for scripts)	chamber p–V curve from a CSV file, energy terms	`calibrate.py` with 10 bench models (motor, pump, pipe, chamber, valve, tail static and dynamic, thrust, hull tow and coast-down, ballast): least squares with 1σ uncertainty and correlation matrix

5 · Comparison

Capability matrix

● modelled in the demo ◐ simplified or partial ○ not modelled

CAPABILITY	MUJOCO	STONEFISH	PYELASTICA	SOFA	OPENFOAM FSI	OPENMODELICA
Free swimming of the whole fish	●	●	●	○ clamped	○ head fixed	◐ surge + heave
Soft tail deformation	◐ 5 segments	◐ 5 segments	◐ continuous rod	● 3D FEM	● 2D FEM	○ 1 DOF
Chamber pressure from structure	◐ constant C_h	◐ constant C_h	◐ constant C_h	●	◐ pressure is input	◐ p–V table
Water flow, wake, vortices	○	○	○	○	●	○
Hydrodynamic forces	◐ ellipsoid model	◐ per mesh face	◐ Morison + Lighthill	◐ drag only	● Navier–Stokes	◐ Lighthill + C_d
Added mass	◐ diagonal	◐ axis-averaged	◐ trailing edge	○	● from the flow	◐ constant
Pump and valve dynamics	◐	◐	◐	◐	○	●
Electrics and energy budget	○	○	○	○	○	●
Buoyancy and depth control	●	●	●	○	○	◐ 1D heave
Noisy sensors	○ possible	●	○	○	○	○
Currents and environment	○	●	○	○	◐ inflow	○
Interactive 3D view	●	●	○ GIF	●	◐ ParaView, after the run	○ plots
Real time or faster	●	● 5–8×	◐ ≈1.5× slower	○ 100–800× slower	○ hours	● milliseconds
Parameter identification	○ plan only	○	○ plan only	○ plan only	○	● calibrate.py
Co-simulation / export	○	◐ ROS 2 possible	○	○	● preCICE	● FMU 2.0

Inputs at a glance

INPUT	MUJOCO	STONEFISH	PYELASTICA	SOFA	OPENFOAM FSI	OPENMODELICA
Geometry	MJCF generated in Python	OBJ meshes + `.scn` XML	station table in Python	gmsh → VTK + OBJ	gmsh → OpenFOAM mesh + CalculiX mesh	none (lumped parameters)
Material	densities, joint stiffness	densities, joint stiffness	E, ν, ρ per element	E, ν, ρ, fibers, spine	E, ν, ρ	tail k, J, c
Hydraulics	lumped (A, r, C_h)	lumped (A, r, C_h)	lumped (A_r, C_h)	volume per chamber	pressure per chamber	component parameters + p–V table
Water	`fluidcoef`	from the mesh + lift slope	C_n, C_t, model choice	C_n, C_t	ρ, ν, inflow U	C_d, C_T, added mass
Config format	Python dataclass	JSON + XML	Python dataclass	Python dataclass + env vars	dictionaries, `.inp`, XML	Modelica parameters
Live control	keyboard	keyboard	none	GUI	none	none

Outputs at a glance

OUTPUT	MUJOCO	STONEFISH	PYELASTICA	SOFA	OPENFOAM FSI	OPENMODELICA
Main quantity	speed, trajectory, depth	speed, trajectory, sensor readings	speed, body shape, thrust	pressure, bend, tethered thrust	flow field, forces, wake	currents, pressures, energy
Data format	CSV	CSV 100 Hz + JSON	CSV	CSV + NPZ	OpenFOAM fields, `.frd`, logs	`.mat`, CSV
Visual	viewer, PNG	3D GUI, PNG	PNG, GIF	GUI, MP4	ParaView, PNG	PNG plots
Live readouts on screen	overlay: state, drive, water force per body; force arrows	panel: state, drive, drag per part, fin lift; library force lines	animation panel: speed, Δp, angles, drag, reactive force; arrows	text: hydraulics, tail, force per slice; arrows; skin coloured by displacement, load or von Mises stress	ParaView text: force on body and tail, tail moment; arrows	OMEdit: plot of any variable after the run

Same fish, different speeds: why

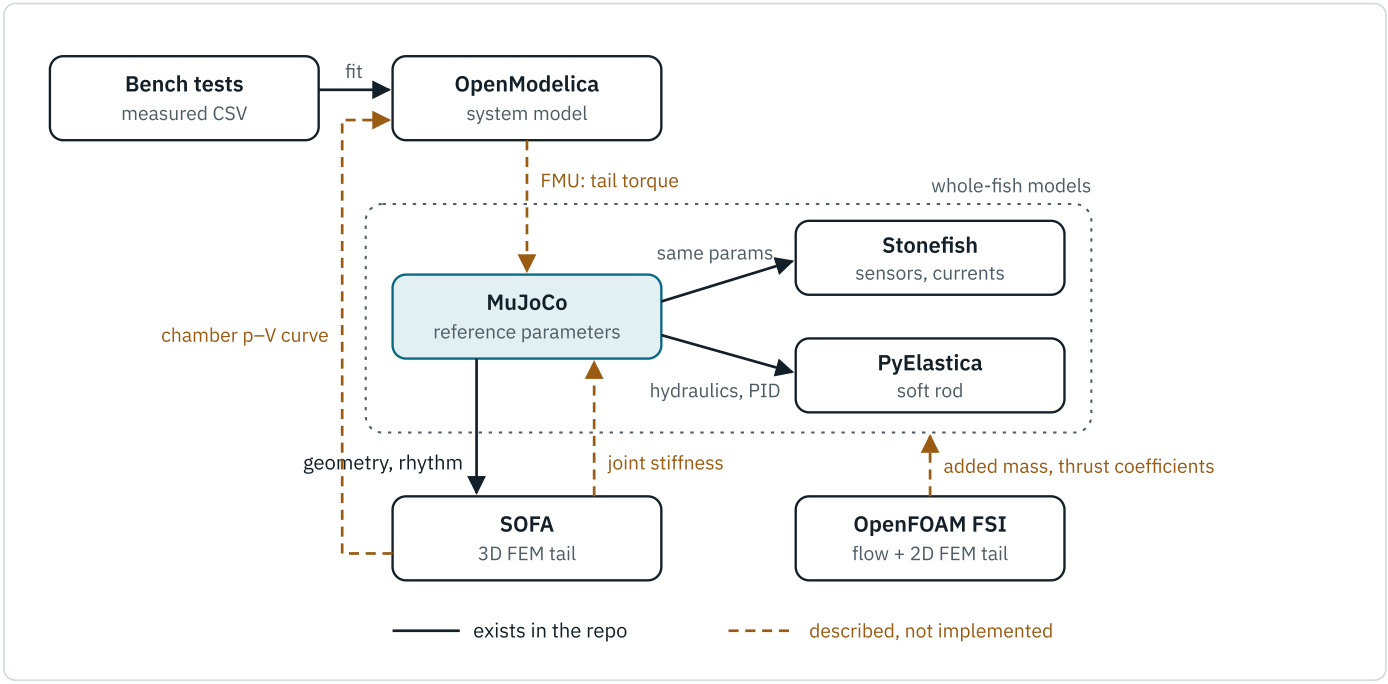
At about 2 Hz the three free-swimming demos give speeds that differ by 2×. The difference comes from where each tool gets its thrust.

TOOL	SPEED NEAR 2 HZ	MAIN SOURCE OF THRUST IN THE MODEL
MuJoCo	22 cm/s	Kutta lift on ellipsoids from the built-in model
Stonefish	34 cm/s	flat-plate fin lift added in C++
PyElastica	44 cm/s	Lighthill reactive force at the trailing edge
OpenModelica	7.3 cm/s	Lighthill rigid fin on a 1-DOF tail (only 3.4° amplitude at 2 Hz)
SOFA	tethered	drag on skin triangles: 63 mN at 2 Hz
OpenFOAM FSI	tethered	resolved flow: net drag with this tail shape

Cost of use

TOOL	INSTALL	TYPICAL RUN	SKILLS NEEDED
MuJoCo	pip, minutes	seconds to minutes	Python
Stonefish	build from source, GPU for the GUI	1–2 min for all scenarios	C++, XML
PyElastica	pip, minutes	9–70 s per stage	Python, rod mechanics
SOFA	script, ≈2 min + compiler for CHOLMOD	10 min to 1 h per stage	Python, FEM, meshing
OpenFOAM FSI	≈1.5 h, three programs and adapters from source	7–102 min per case	CFD, FEM, coupling
OpenModelica	apt repository + venv	under a second	Modelica

How the tools fit together



MuJoCo is the reference: Stonefish reuses its fish and parameters, PyElastica ports its hydraulics and depth PID, and SOFA copies its tail geometry and rhythm. The dashed links are the planned next steps, each described in a README: SOFA p–V curves into the OpenModelica chamber table, the OpenModelica FMU driving a MuJoCo joint, SOFA/PyElastica stiffness exported to MuJoCo joints, and OpenFOAM results used to calibrate the cheaper water models.

6 · Glossary

ENGLISH	POLISH	MEANING HERE
added mass	masa dodana	water that has to be accelerated together with the body; in the FSI demo it lowers the tail resonance from 3.39 Hz to ≈1 Hz
corotational FEM	MES korotacyjna	linear elasticity that handles large rotations (SOFA)
Cosserat rod	pręt Cosseratów	1D continuum that bends, twists, shears and stretches (PyElastica)
CPG (rhythm generator)	generator rytmu	controller that produces the periodic volume command for the tail
FMU	FMU (Functional Mock-up Unit)	packaged model that other tools can run (OpenModelica export)
FSI	interakcja płyn–struktura	coupled simulation of water flow and a deforming solid (OpenFOAM + CalculiX)
identification	identyfikacja parametrów	fitting model parameters to bench measurements (OpenModelica `calibrate.py`)
IQN-ILS	IQN-ILS (quasi-Newton)	method preCICE uses to make implicit coupling converge
Lighthill reactive force	siła reakcyjna Lighthilla	thrust from the water pushed sideways by the trailing edge
PRBM	model pseudo-sztywny	soft tail approximated by rigid segments and springs (MuJoCo, Stonefish)
p–V curve	krzywa ciśnienie–objętość	chamber pressure as a function of the water volume pumped in
relief valve	zawór przelewowy	valve that opens at $\|\Delta p\|$ = 50 kPa to protect the chambers
STEP	STEP	neutral CAD exchange format (ISO 10303); gmsh imports it directly
Strouhal number	liczba Strouhala	$St = f \cdot A/U$; efficient swimmers cruise at $St \approx 0.2–0.4$, most fish 0.25–0.35 (Taylor et al. 2003; Triantafyllou et al. 1993)
tethered	na uwięzi	body held in place while thrust is measured (SOFA, OpenFOAM)
VBS	układ zmiennej pływalności	variable buoyancy system: a tank whose water volume sets depth (Stonefish)
watertight mesh	siatka szczelna (zamknięta)	closed surface with no holes; needed for buoyancy in Stonefish and cavities in SOFA

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